

BOWEN (“BEE”) BOTHELLO

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SUMMARY OF QUALIFICATIONS

Software Engineer and interactive designer with experience in C#, C++, Python, Unity Engine, ROS 2, and real-time interactive system development. Skilled at building gameplay systems, state machines, virtual reality experiences, and hardware-integrated attractions that combine technical reliability with engaging user experiences. Strong collaborator who works effectively with engineers, artists, designers, fabricators, and researchers to transform creative ideas into interactive functional experiences. Proven ability to lead teams, manage complex projects, and develop scalable software solutions across games, immersive experiences, research, and entertainment applications.

EDUCATION

Bachelor of Science in Computer Science | University of California, San Diego | September 2024 – September 2026

Relevant Coursework: Software Engineering, Human Robot Interaction (HRI), Operating System Principles, Design and Analysis of Algorithms, Human Computer Interaction (HCI), and Introduction to Artificial Intelligence

TECHNICAL SKILLS

Programming Languages: C#, C++, Python, C, Java, JavaScript

Software: Unity Engine, ROS 2, RocksDB, Git, Github, Arduino IDE, Raspberry Pi, NoLimits 2, Trello, Slack, Discord, Adobe Creative Cloud, Microsoft Office, Google Suite, Zoom

Highlights: Game development, multi-threaded system development, hardware integration, virtual reality development, multi-disciplinary collaboration, product management, object-oriented design, data science, machine learning principles

RELEVANT EXPERIENCE

Project T.O.T.T.Y | *Integration Engineer, Gameplay Systems Designer* | May 2026 – June 2026

- Developed a real-time interactive system on the ROS 2 software library within a Raspberry Pi attached to the *Pupper* robot with sensor integration including the OAK-D Lite depth camera for OpenCV computer vision.
- Designed and implemented a practical game state machine architecture to manage the progression of a unique game of competitive “Would You Rather” utilizing the *Pupper* as the embodied agent host for a group of four participants.
- Built an adaptive experience which responds dynamically to the input of players and balanced to subtly maintain a close score between participants through conditional audio outputs.
- Conducted focus groups and post-experimental interviews to refine the overall experience of the project.
- Researched Human Robot Interaction concepts from group robot dynamics to gaze response to create a robot design which can lead a group as an embodied host throughout the gameplay cycle.

Cosmic Swirl | *Electrical Team Lead, Integration Engineer, Electrical Designer* | January 2026 – April 2026

- Managed a team of electrical-focused teammates throughout the project to ensure steady and consistent development each week while competing in the *Ride Engineering Competition*.
- Developed a consistent, reliable ride state management system and ride controller, capable of taking in real-time button inputs from the ride’s control panel within microseconds and dynamically responding accordingly without endangering the potential riders.
- Integrated multiple Arduino microcontrollers into the system which communicated to each other through an I2C communication protocol.
- Implemented physical components from motors, servos, to LED strips for the ride which were all synchronously controlled through the centralized ride state machine logic.
- Collaborated cross-functionally with engineers, artists, and fabricators to translate creative ride concepts into technical solutions.
- Led the on-site installation and testing of the core microcontroller system at the competition site.

Starbright Travelers | *Software Engineer, Immersive Experience Designer, Product Manager* | January 2026 – April 2026

- Implemented several object-based scripts into the No Limits 2 virtual ride simulation utilizing the unique *nolimitscoaster* API to visualize the design of kinetic set pieces for the competition *Swamp Thrills*.
- Collaborated with a team of several disciplines from conceptual artwork to CAD modeling to translate the creative concepts into technical systems.
- Maintained a consistent and cohesive vision for the project and communicated frequently with the team members to ensure a stable and well-scoped product which was developable within the competition timeline.

MetaNomads Research System Development | *Gameplay Programmer, Scrum Master* | April 2025 – April 2026

- Led a team on the gameplay programming track focused on the continued development and implementation of the *MetaFrame* Unity Engine real-time interactive virtual reality experimental data tracking and recording package in the U.C. San Diego Cognitive Sciences XR Lab.
- Integrated a Scrum workflow to ensure consistent progress from all team members each week.
- Designed and implemented data collection tools for the precise tracking of independent eye gaze and head movements.
- Built scalable and modular systems for gaze and hand interaction in the VR space to enable ease of event implementation for researchers.
- Translated storyboarded interaction events from the research director into technical gameplay systems.
- Developed a core state machine for progressing the experimental tests through the various interaction settings necessary for the research.

Adaption of Game Shoulders of Giants | *Gameplay Programmer, UI/UX Designer* | September 2025 – December 2025

- Developed the Unity Engine digital adaption of a scalable and modular trading card game system using object-oriented principles and intuitive interface practices to allow for expansion from non-programmer developers.
- Crafted a user experience of highly responsive and fluidly moving buttons and cards tailored to be engaging for the key demographic of young students learning about history.
- Translated the creative briefs from the card game designer into technical system architecture.
- Created an adaptive system capable of responding dynamically to the players’ thousands of potential card placement options.

Web-based US Exploration Game *Geoquest* | *Systems Designer, Scrum Master* | September 2025 – December 2025

- Led a group of 6 developers to create a successful interactive website utilizing Typescript and Konva by following Agile methodologies and practicing the role of scrum master.
- Designed an object-focused backend system for ease of integration for other developers to display information of various cities for the key stakeholder of students in elementary and middle school.
- Upheld and maintained a protected Github repository as the designated Git Lead of the project through proper permission management to reduce the potential for merge conflicts and improve the ease of implementation.
- Implemented bug detection methods such as automated unit testing to ensure a consistent and stable main branch for the project codebase.

Multithreaded Synchronization Research | *Systems Designer, Software Analyst* | September 2024 – June 2025

- Designed and implemented applications focused on benchmark testing a new low-level synchronization primitive method under diverse, real-world database conditions.
- Analyzed experiment data tracking low tail latency and CPU efficiency in a multithreaded system and compared the results to existing solutions. [Research poster](#).

Commercial Video Game *Deitrus* | *Gameplay Designer, Gameplay Programmer* | February 2019 – August 2024

- Led a team of three game developers to ship *Deitrus*, a narrative-driven action-adventure RPG released on Steam and Itch.io, and showcased at [South by Southwest \(SXSW\)](#).
- Developed state machines and event-driven architectures to manage the game's logic and narrative progression in the real-time gameplay environment.
- Crafted intuitive, object-oriented tools within Unity Engine to improve ease of creative concept integration into the project for other team members.
- Conducted user playtesting sessions and utilized the feedback data to refine the gameplay system experience.
- Optimized game performance to retain a low latency and at minimum 30 FPS processing for most standard computer systems on both Windows and Mac operating systems.

ADDITIONAL EXPERIENCE

Public Transit Enthusiasts @ U.C. San Diego | *Operations Director, Event Coordinator* | February 2025 – June 2026

- Improved the operations of the organization by introducing a consistent officer meeting schedule with a structure focused on forward growth and iterative improvement.
- Allocated weekly tasks for the officers to handle between meetings and maintained consistent progress through frequent communication and support when necessary.
- Designed unique, transit-oriented events to continually entice and engage members which raised retention by over 50% from what it was before elected officer.

Haunt Comp Project | *Project Manager, Immersive Environment Designer* | February 2026 – May 2026

- Led a multidisciplinary team of 6 members to design and visualize a conceptual haunted attraction for the competition *Haunt Comp* by leading meetings, utilizing Scrum principles, and maintaining proper spreadsheet management.
- Designed a terrifying haunted house environment through the careful placement of lighting, speakers, and technical devices within the layout, and created a theoretical SOP to properly manage such an experience.
- Handled unexpected obstacles from team members through professional communication, timeline restructuring, scope management, and task reallocation.